Indroneel Roy

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Research Interests: *Medical Image Segmentation · Hybrid CNN-Transformer Architectures · Efficient Deep Learning for Clinical Applications · Computer Vision*

Research Summary

Computer vision researcher and EEE graduate from SUST, focused on Transformer-based architectures and LLMs for visual tasks. From hybrid CNN-Transformer segmentation to ViT-based classification, my work centers on building smarter, more efficient vision systems. Seeking a PhD to take this further.

Education

Shahjalal University of Science and Technology (SUST) February 2020 – August 2025
B.Sc. in Electrical & Electronic Engineering ***GPA: 3.22/4.00***

– **Capstone Focus:** Hybrid CNN-Transformer architectures for medical image segmentation

Research Projects

MSHFNet: Multi-Scale Hybrid Fusion Network for Multi-Organ CT Segmentation | ***PyTorch, timm*** GitHub ↗

– Proposed MSHFNet, a dual-pathway encoder-decoder architecture integrating a pretrained ResNet-50 CNN encoder with a Swin-Base Transformer encoder, designed for precise multi-organ segmentation on abdominal CT scans.

– Designed a novel Cross-Attention Fusion Module (CAFM) performing bidirectional cross-attention at four hierarchical scales, enabling each modality to selectively enrich the other beyond simple concatenation or addition.

– Implemented a Shared Decoder with Deep Supervision using a combined Cross-Entropy, Dice, and Boundary loss, providing auxiliary gradient signals at three intermediate resolutions to improve boundary-aware learning.

– Achieved **79.73%** mean DSC on the Synapse multi-organ benchmark (8 organs), surpassing TransUNet by +2.25% DSC, with state-of-the-art Pancreas segmentation at **81.47%**; further achieved **91.18%** mean DSC on the ACDC cardiac MRI benchmark, confirming cross-domain generalisability.

PolyFormer: Hybrid CNN-Transformer Network for Polyp Segmentation | ***PyTorch, OpenCV*** GitHub ↗

– Designed a novel encoder-decoder architecture integrating a pretrained **ResNet-50 CNN** backbone with a custom **Transformer bottleneck** featuring learnable positional encodings and 8-head multi-head self-attention for global context modeling in colonoscopy images.

– Developed a **Cross-Attention Fusion Module** where Transformer features attend to CNN spatial features as key-value pairs, enabling explicit synergy between local texture representations and long-range spatial dependencies.

– Implemented a **Combined Dice + BCE loss** with a U-Net style hierarchical decoder using four skip-connection stages, improving boundary precision for small and irregularly shaped polyps.

– Achieved **Dice: 0.8971** and **IoU: 0.8172** on the **Kvasir-SEG** benchmark (60 epochs, NVIDIA T4 GPU), trained with AdamW optimizer and Cosine Annealing scheduler.

Cassava Leaf Disease Classification using Vision Transformer (ViT) | ***PyTorch, timm*** GitHub ↗

– Designed and fine-tuned a ViT-Base/16 model for multi-class plant disease classification across 5 categories on the Kaggle Cassava Leaf Disease dataset (21,367 training images).

– Achieved **87.15%** validation accuracy with **93.4%** precision and **95.6%** recall for Cassava Mosaic Disease (CMD), outperforming ResNet-50 baseline by ∼4.2%.

– Implemented full training pipeline with transfer learning via timm, GPU-accelerated 224×224 preprocessing, label smoothing, and cosine LR scheduling.

Vision-Based Wheat Head Detection using YOLOv11 | ***PyTorch, OpenCV*** GitHub ↗

- Fine-tuned YOLOv11-Medium on a large-scale wheat head detection dataset with **29,422+** annotated instances for precision agriculture applications.
- Achieved **94.7% mAP@50** and **55.4% mAP@50–95** with **91.7%** precision and **89.2%** recall, demonstrating strong generalization under varying occlusion and lighting.

Agentflow: Multi-Tool Conversational AI System | ***LangChain, LangGraph, LLM*** GitHub ↗

- Designed a stateful conversational AI system using LangGraph state machine architecture with persistent SQLite checkpointing for robust multi-turn dialogue management.
- Integrated 10+ specialized tools including web search, financial APIs (Alpha Vantage, CoinGecko), and weather services with dynamic tool binding using Groq LLM (Llama-3.3-70B).

Research & Model Implementations

- **CNN Architectures** — Implemented and trained LeNet, AlexNet, ResNet, DenseNet, and EfficientNet from scratch in PyTorch; performed architectural comparisons and benchmarked on standard image classification tasks.
- **Transformer-Based Models** — Implemented Transformer, Swin Transformer, and TransUNet; investigated self-attention mechanisms, window-based attention, and hierarchical vision representations for classification and segmentation.
- **Image Segmentation Models** — Implemented U-Net, SegFormer, and MaskFormer; conducted experiments on medical and natural image segmentation datasets using Dice and IoU metrics.
- **Code Repository** — All implementations publicly available and reproducible on GitHub ↗.

Technical Skills

Languages: Python, SQL, C, MATLAB, Assembly (Intel ×86)
ML/DL Frameworks: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV
Computer Vision: CNNs, Object Detection (YOLO), Image Segmentation, ResNet, U-Net, ViT, Swin Transformer
LLM & NLP: LangChain, LangGraph, Transformers, Hugging Face, RAG, Vector Databases
MLOps & Deployment: Docker, FastAPI, Streamlit, Hugging Face Spaces, Git, AWS
Databases: SQL, Pinecone, ChromaDB
Algorithms & DS: Data Structures & Algorithms 200+ LeetCode problems solved

Certifications

Neural Networks and Deep Learning — Coursera / deeplearning.ai (View Certificate)

Additional Information

Languages: English (Fluent), Bengali (Native)

Interests: Open-source AI contributions, research paper implementations, technical blogging, chess